

Planning in the LLM Era: Building for Reliability and Efficiency

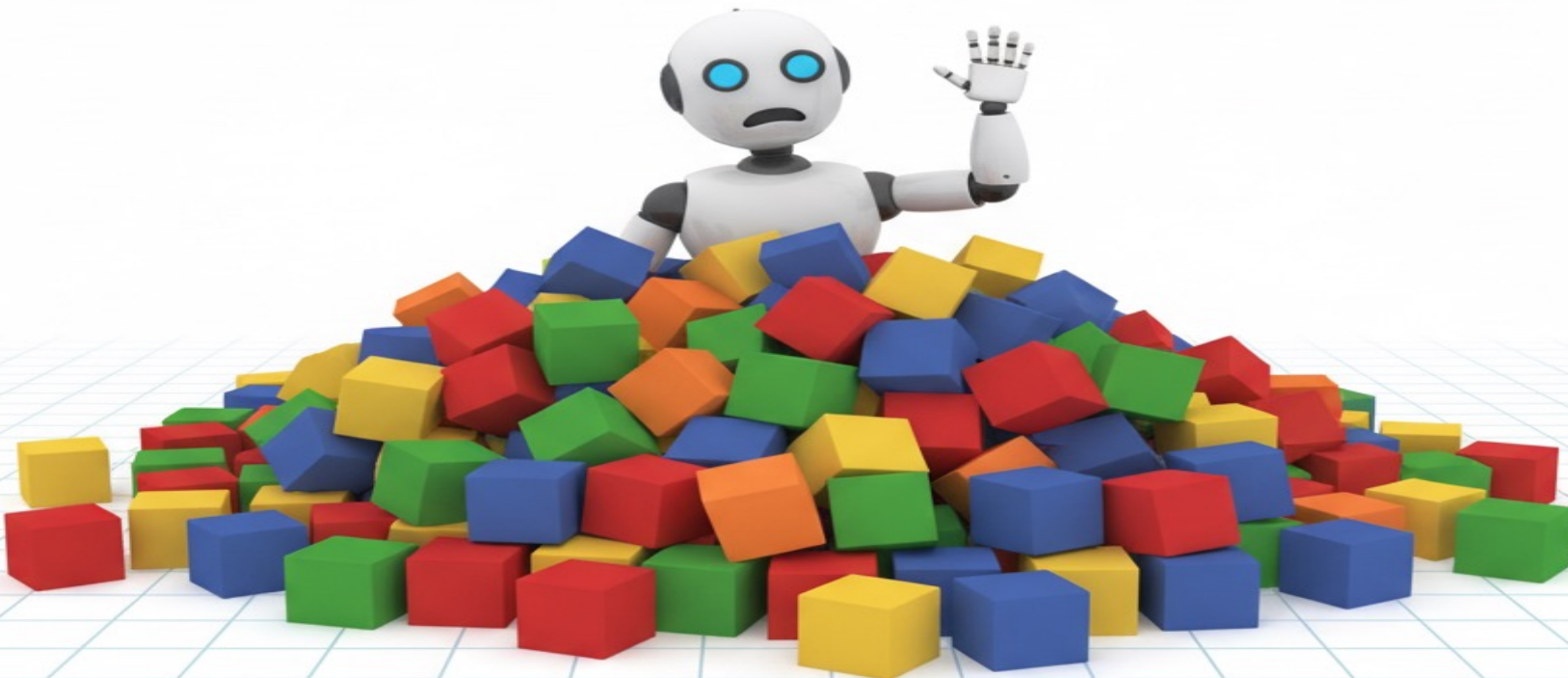
Michael Katz

Harsha Kokel

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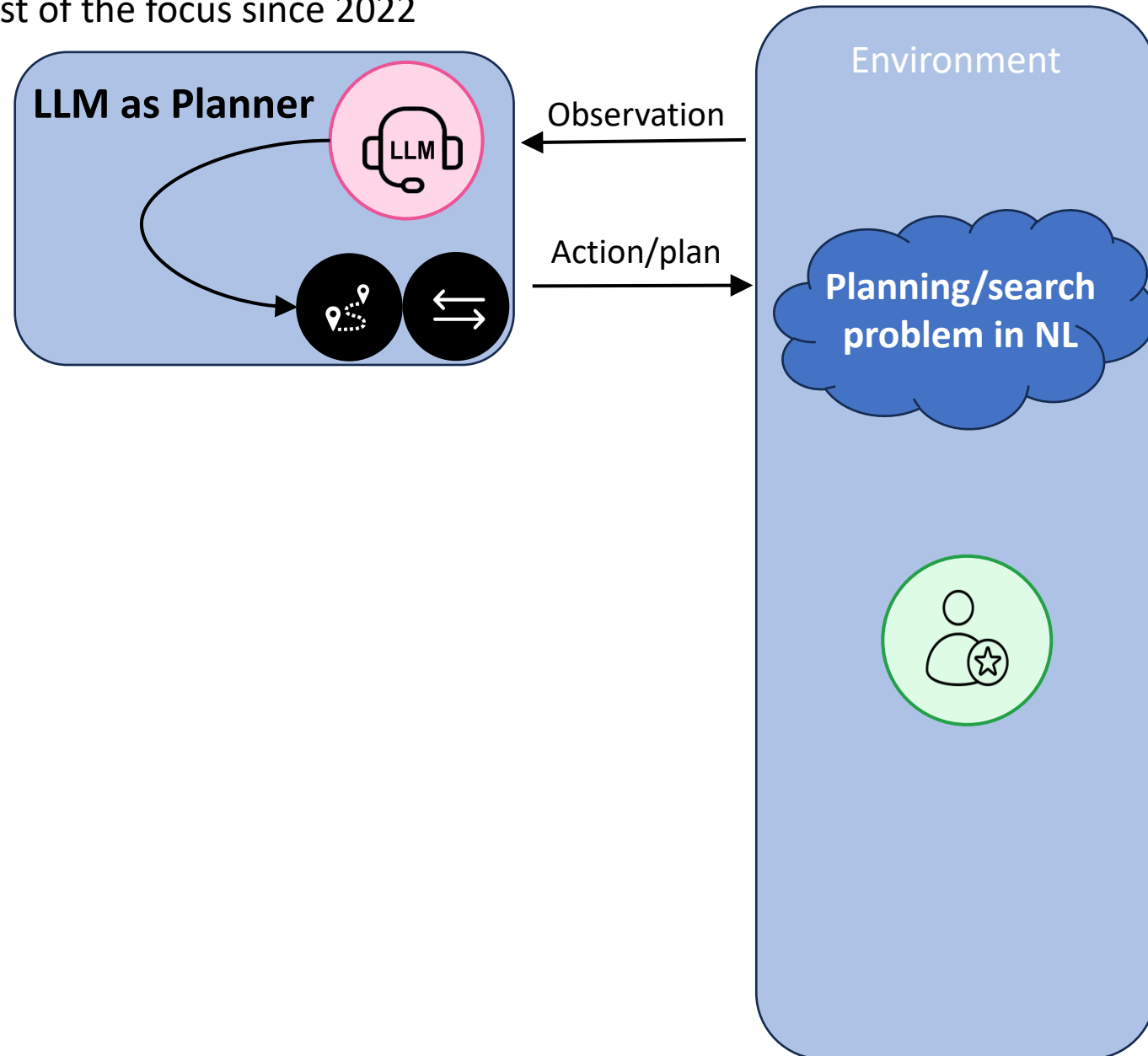
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Motivation

Most of the focus since 2022



- Planning is central to intelligent agents
- LLMs sparked renewed interest
- What should we do with them?

Why LLM as Planner Falls Short

- Unsound and incomplete methods
- High computational cost per task
- No reuse across tasks
- Fragile to change



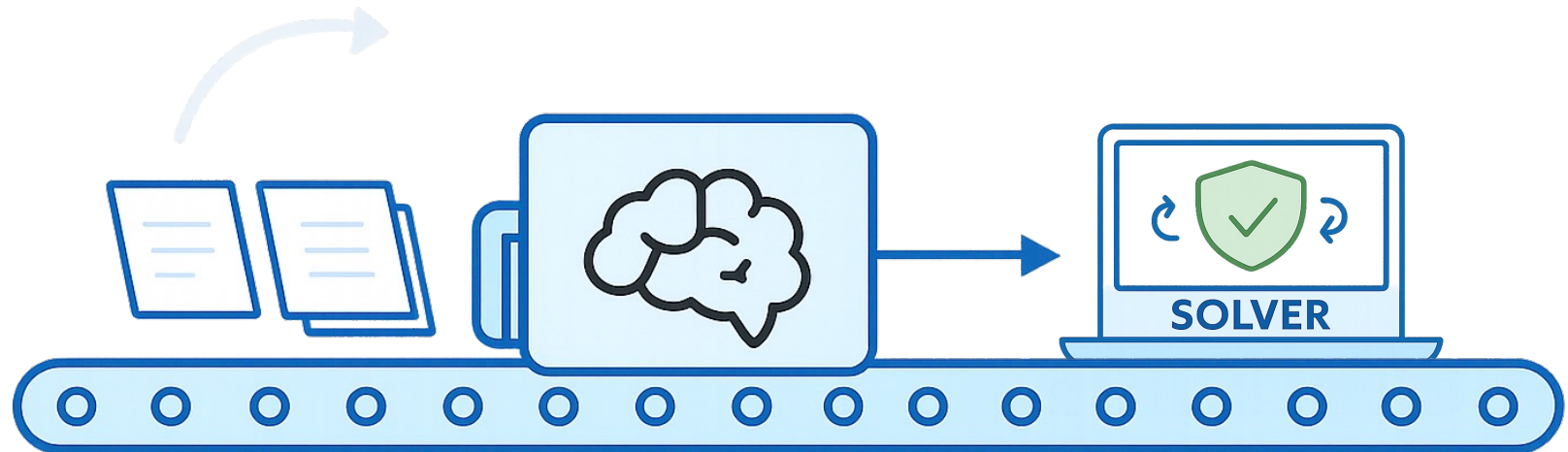
Central Idea: LLM as Solver Generator

Use LLMs mostly at construction time, not inference time

- Generate a solver
- Validate
- Reuse efficiently

Benefits:

- Reliability
- Efficiency
- Maintainability



PLANNING IN THE MODERN LLM ERA



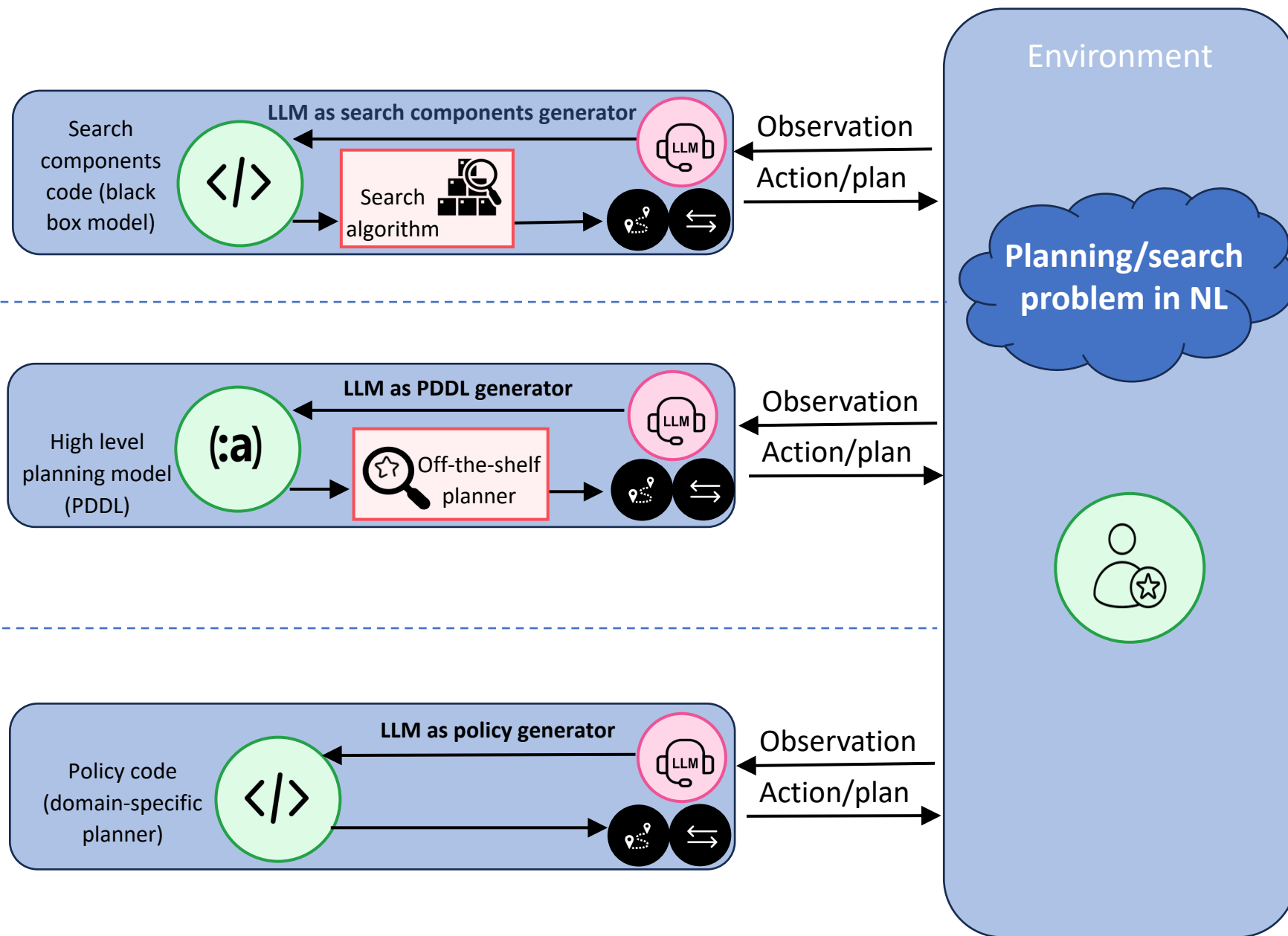
The Three Paradigms

Katz et al, NeurIPS 2024

Cao et al, ICAPS 2026

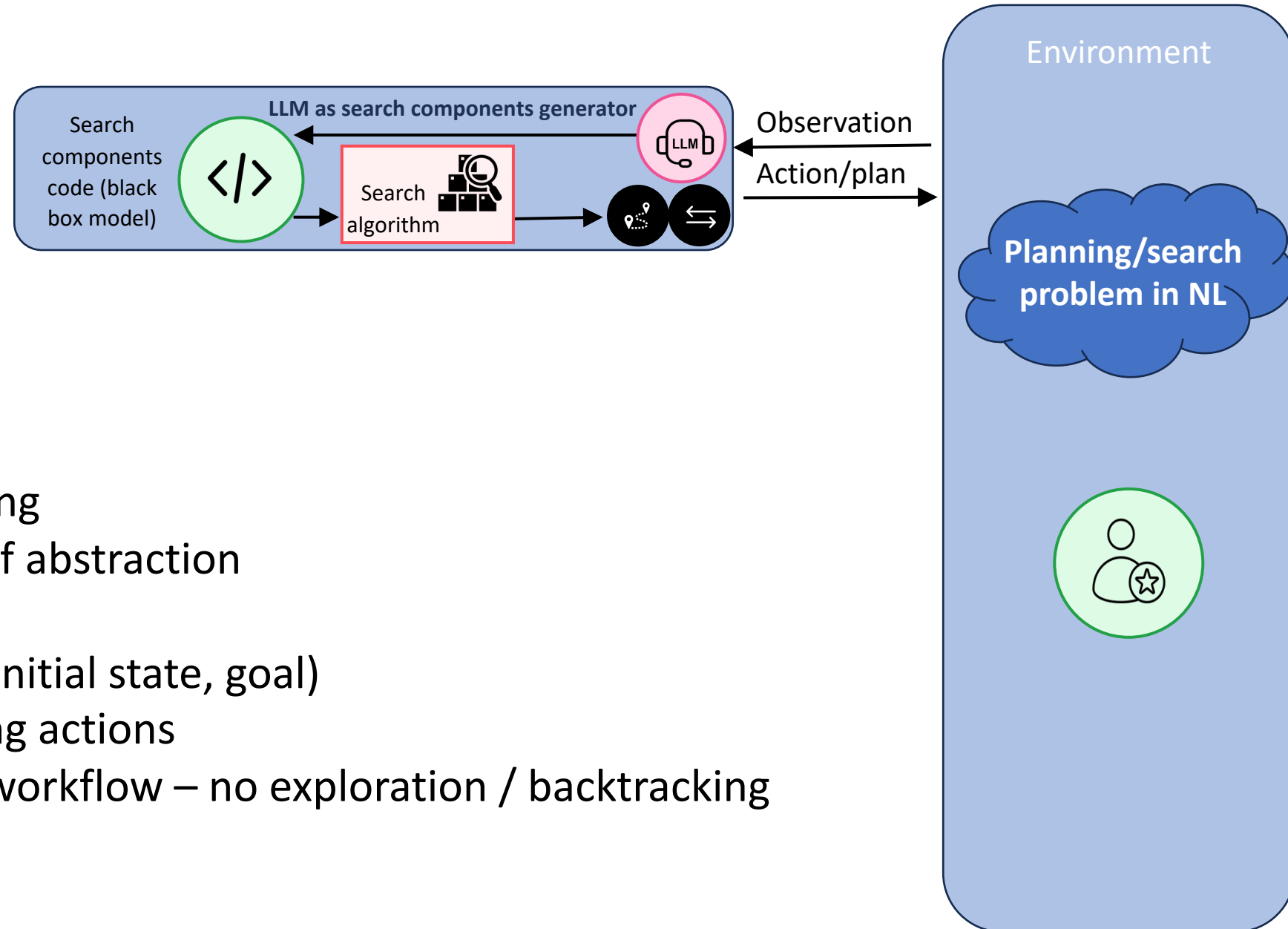
Tuisov et al, ICAPS 2026

Correa et al, NeurIPS 2025



Search Components Generation

Katz et al, NeurIPS 2024
Cao et al, ICAPS 2026
Correa et al, NeurIPS 2025
Tuisov et al, ICAPS 2026

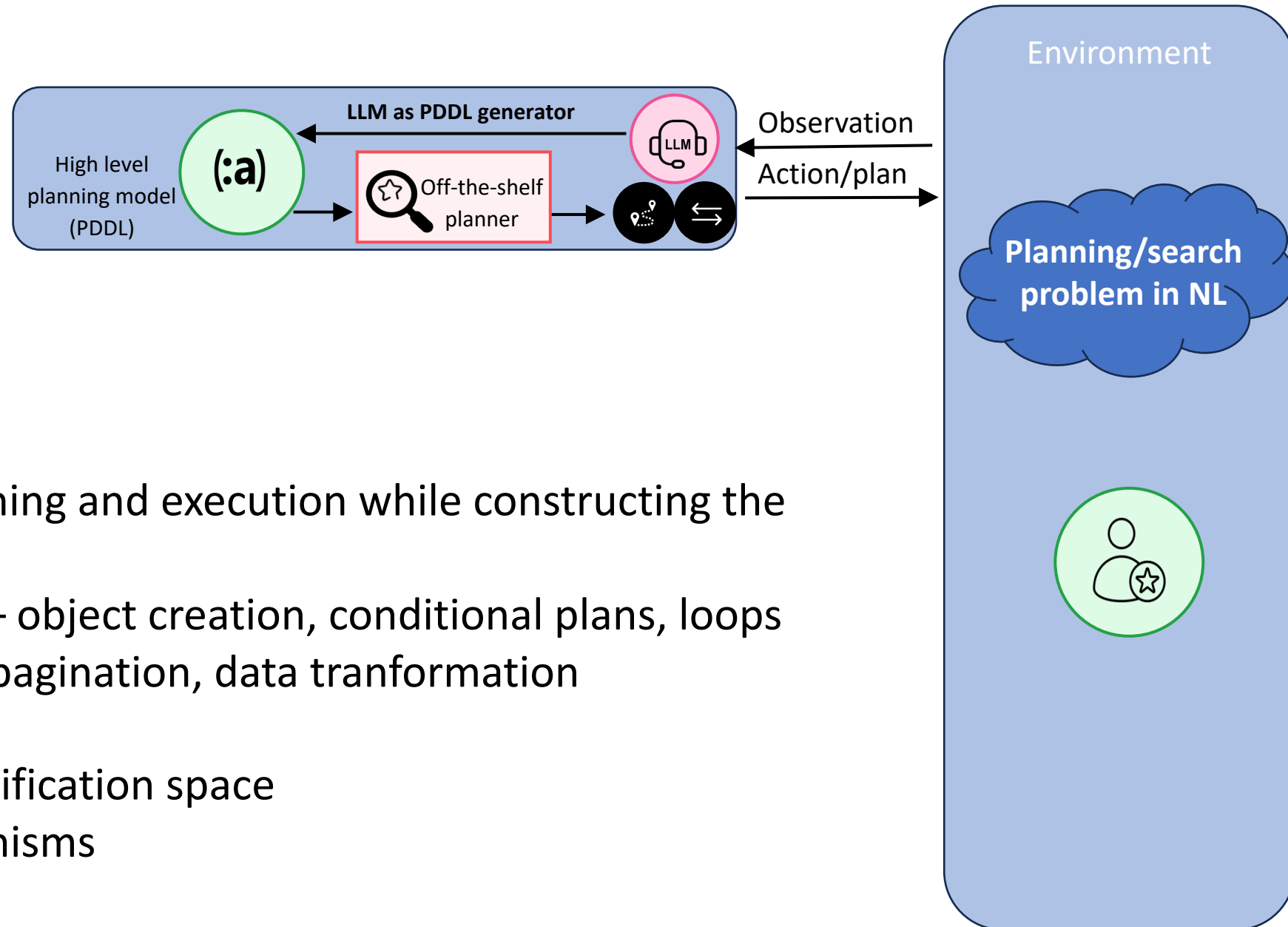


Open challenges:

- State representation
 - State features learning
 - Find the right level of abstraction
- Partial information
 - Hidden knowledge (initial state, goal)
 - Information gathering actions
- Linear implementation workflow – no exploration / backtracking
- Validation required

PDDL Generation

Guan et al, NeurIPS 2023
Gestrin et al, HAXP@ICAPS 2024
Oswald et al, ICAPS 2024
Huang et al, AAI 2025
Tantakoun et al, ACL Findings 2025
Oswald et al, WM@ICLR 2026



Open challenges:

- Same as before
- Interplay between planning and execution while constructing the planning model
- Non-standard features – object creation, conditional plans, loops
- Procedural glue code – pagination, data transformation
- PDDL evaluation
- Search in the PDDL modification space
- Richer feedback mechanisms
- Validation required

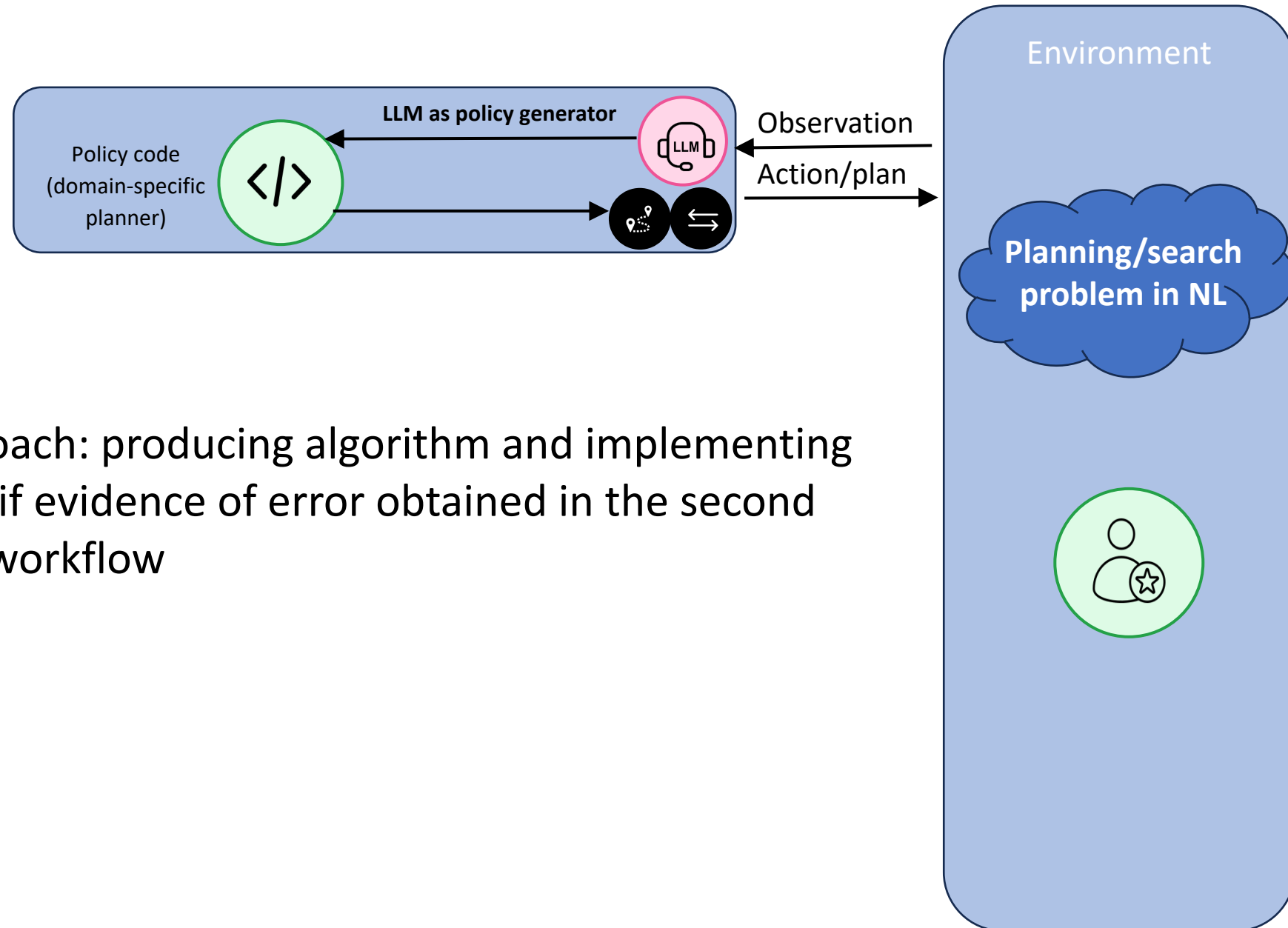
Generalized Policy Generation

Silver et al, AAI 2024

Hodel, BSc Thesis 2024

Stein et al, ICAPS 2026

Sohrabi et al, LM4Plan@ICAPS 2026



Open challenges:

- Conceptual 2-step approach: producing algorithm and implementing
 - Revisit the first step if evidence of error obtained in the second
- Linear implementation workflow
- Validation required

Cross-Category Observations

Complementarity of methods

- Can improve each other: e.g., transition function code can help generate PDDL



Advantages of Policy Generation

- Procedural glue code natively supported
- Loops, conditions, etc.
- Interleaving sensing with acting is more natural
- Reusable skills and hierarchical structure
- Partial observability

Conclusions

Use LLMs mostly at construction time, not inference time

- Three emerging directions:
 - NL2Search
 - NL2PDDL
 - NL2Policy
- All show promise, but have limitations

Takeaway:

- Ensure producing planners that are efficient, reliable, and able to handle the required problems

QUESTIONS?

